WW-P-401E
October 24, 1974
SUPERSEDING
Fed. Spec. WW-P-401D
April 30, 1969

FEDERAL SPECIFICATION

PIPE AND PIPE FITTINGS, CAST-IRON, SOIL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cast-iron soil pipe and .ttings.

1.2 Classification.

1.2.1 Types, classes, patterns, and sizes. The cast-iron soil pipe id fittings covered by this specification shall be furnished in the)llo~ing types, classes, patterns, and sizes, as specified (see 6.2):

Type I - Single hub (applies to pipe only).
Type II - Double hub (applies to pipe only).
Type III - Hubless cast-iron sanitary system.

Class XH - Extra heavy.
Class SV - Service weight.

Patterns - Various, to be selected from ASTM A74 or CISPI 301 (applies to fittings only).

Sizes - Various, to be selected from ASTM A74 or CISPI 301.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation or bids or request for proposal form a part of this specification to the xtent specified herein:

FSC 4710
4730

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Federal Specifications:

QQ-S-781	- Strapping, Steel, Flat and Seals.
PPP-B-587	- Boxes, Wood, Wirebound Pallet Type.

Federal Standard:

FED. STD. No. 123	- Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of Military Standards required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials (ASTM) Standard:

A74 - Cast Iron Soil Pipe and Fittings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Cast Iron Soil Pipe Institute (CISPI):

CISPI 301 - Hubless Cast Iron Soil Pipe and Fittings.

(Application for copies should be addressed to Cast Iron Soil Pipe Institute, 2029 K Street, NW, Washington, DC 20006.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The cast-iron soil pipe and fittings shall be in accordance with ASTM A74 or CISPI 301, and as specified herein.

3.2 Materials. Materials shall be as specified in ASTM A74 or CISPI 301.

3.3 Coating. Unless otherwise specified (see 6.2), the pipe and fittings shall be coated in accordance with ASTM A74 or CISPI 301.

3.4 Identification marking. The pipe and fittings shall be identified in accordance with ASTM A74 or CISPI 301.

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3.5 Workmanship. The pipe and fittings shall be sound, true to pattern, and of compact close grain which permits drilling and cutting. The interior surface shall be smooth and free from defects which would make the castings unfit for the use intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Lot. For the purpose of inspection, a lot shall consist of all pipes and fittings of the same type, class, pattern, and size offered for delivery at one time.

4.3.2 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Level II for examination and Level S-2 for tests.

4.3.3 Examination. Samples selected in accordance with 4.3.2 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.4 Tests. Samples selected in accordance with 4.3.2 shall be tested as specified in 4.4.2. AQL shall be 2.5 percent defective.

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4.4 Inspection procedure.

4.4.1 Examination. Pipe and fittings shall be examined as specified herein, for the following defects:

Major

101. Materials not as specified herein.
102. Coating not as specified.
103. Type not as specified (pipe only).
104. Class not as specified (pipe only).
105. Pattern not as specified (fittings only).
106. Size not as specified.
107. Pipe laying length not as specified.
108. Workmanship not as specified.

Minor

201. Coating on threaded openings not as specified.
202. Marking not as specified.

4.4.2 Tests. Pipe and fittings shall be tested as specified in ASTM A74 or CISPI 301 as applicable. Nonconformance to any requirement of ASTM A74 or CISPI 301 as applicable, shall constitute failure of this test.

4.4.2.1 Thickness. Measure thickness of metal at or near the ends of pipe and fittings by one of the following methods:

(a) Ultrasonic resonant frequency test unit. Calibrate the unit prior to use by measuring the thickness of a coated specimen of known metal thickness.
(b) Transfer-type caliper. Remove coating in way of measurement. When thickness determinations are complete, recoat the areas of measurement.

Thickness not falling within the tolerances specified in ASTM A74 or CISPI 301 as applicable, shall constitute failure of this test.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.5.1.2 Lot. A lot shall consist of all completed packs of any one level offered for examination at one time.

4.5.1.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. AQL shall be 2.5 percent defective.

4.5.1.4 Examination. Samples selected in accordance with 4.5.1.3 shall be examined for the following defects:

109. Materials, methods, and containers not as specified for Level A. Each incorrect material, method, or container shall be considered one defect.
110. Strapping not zinc-coated for Level A.
111. Marking illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be Level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Pipe fittings. The pipe fittings shall be packed in a box conforming to PPP-B-587, Type I, Class 1.

5.1.1.2 Pipe. The pipe shall be blocked, braced, and packed in accordance with Figures 1, 2, and 3. Strapping shall be zinc-coated and shall conform to QQ-S-781.

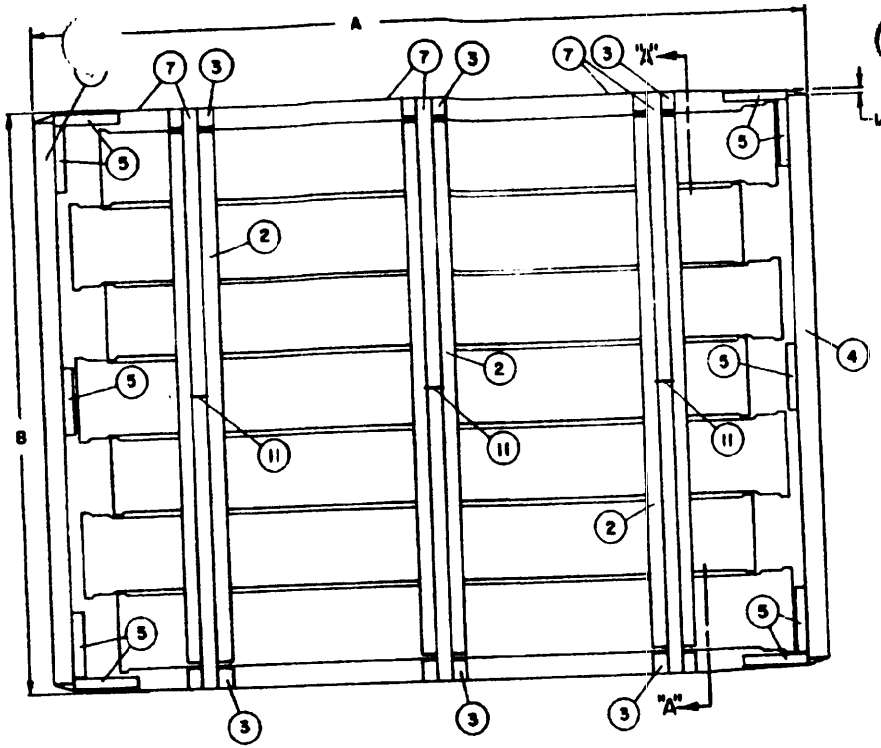
5.1.2 Level C. The pipe and pipe fittings shall be packed or prepared for shipment in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

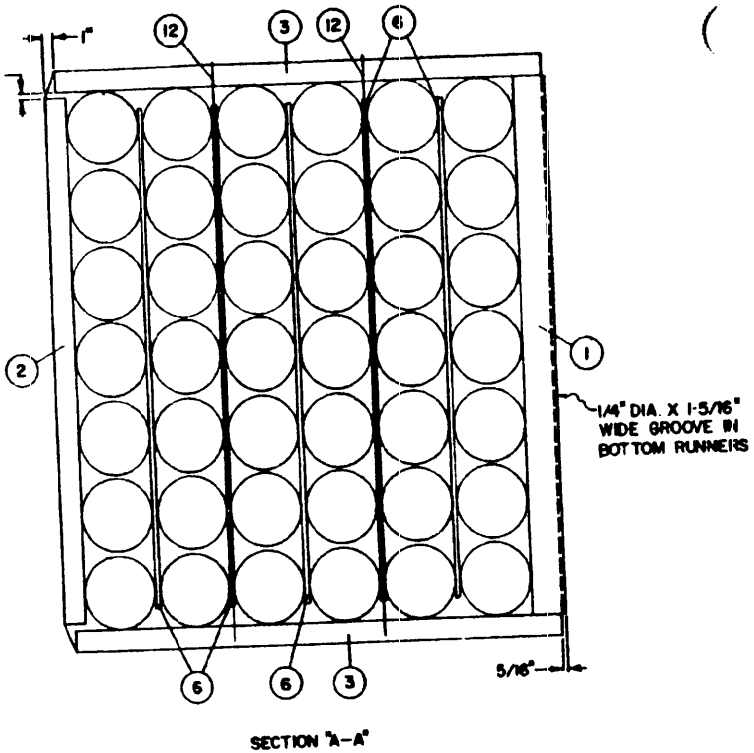
5.2 Marking. Marking shall be in accordance with 5.2.1 or 5.2.2, as specified (see 6.2).

5.2.1 Civil agencies. In addition to any special markings required by the contract or order, all containers shall be marked in accordance with FED. STD. No. 123.

5.2.2 Military agencies. In addition to any special markings required by the contract or order, all containers shall be marked in accordance with MIL-STD-129.

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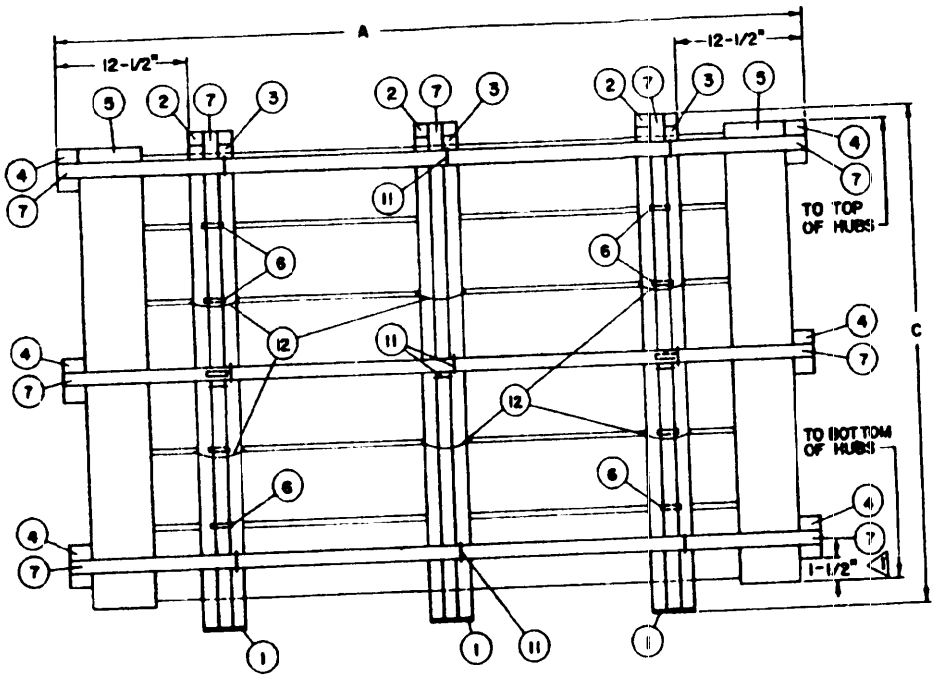


NOTE: SEE TABLES I AND 2 FOR DIMENSIONS AND MATERIAL.

Figure 1. PACKING, LEVEL A.

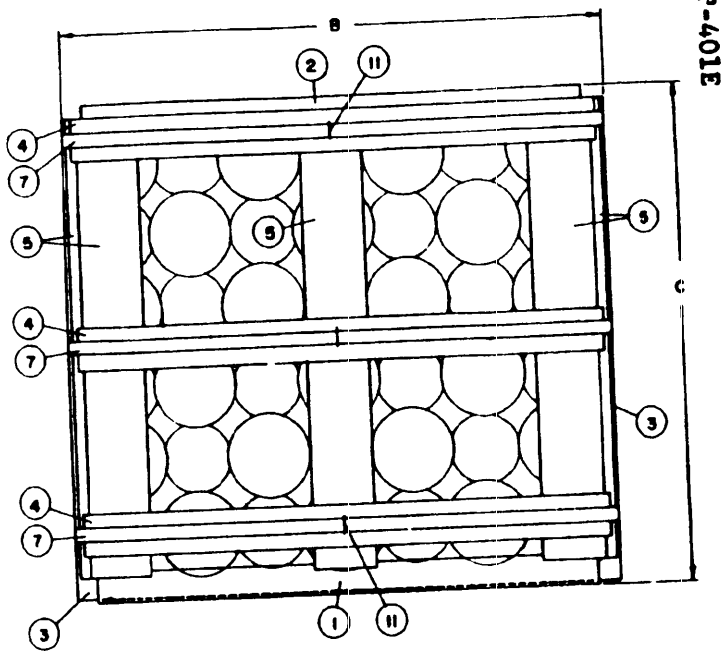
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SIDE ELEVATION

NOTE: SEE TABLES I AND 2 FOR DIMENSIONS AND MATERIAL.



END ELEVATION

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Figure 2. PACKING, LEVEL A.

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PIPE SIZE	TYPE	PIPE / BNDL	ROWS	LAYERS	NOM. PIPE WT. BNDL	O/A DIM. A	O/A DIM. B	O/A DIM. C	MATERIALS INFO M ⚠	N ⚠	P ⚠	R	S ⚠	T	W ⚠
2	SV-SH	196	14	14	3920	71-1/2	38	4.	33-1/2	41-3/4	35-3/4	39-1/2	32-1/4	26	34
2	XH-SH	156	13	12	3900	71-1/2	37-1/4	39	32-3/4	37-3/4	35-1/4	35-1/2	31-1/2	22	33-1/4
3	SV-SH	132	12	11	3960	72	45	45-3/4	40-1/2	44-1/2	43	42-1/4	39	20	41
3	XH-SH	90	10	9	4050	72	40-3/4	40-1/2	36-1/4	39-1/4	38-3/4	37	34-3/4	16	36-3/4
4	SV-SH	100	10	10	4000	72-1/2	48-1/4	52	43-3/4	50-3/4	46-1/4	48-1/2	41-3/4	18	44-1/4
4	XH-SH	64	8	8	3840	72-1/2	41-1/4	44-1/2	36-3/4	43-1/4	39-1/4	41-1/2	35-3/4	14	37-1/4
6	SV-SH	56	8	7	3640	72-1/2	55-1/4	51	50-3/4	50-1/2	53-1/4	48-1/4	47-3/4	12	51-1/4
6	XH-SH	42	7	6	3990	72-1/2	50-3/4	46-1/2	46	45-1/4	48-3/4	43	43	10	46-1/2
8	SV-SH	25	5	5	2500	75	47	49-1/2	42-1/2	49	46	46-1/2	39	8	43
8	XH-SH	25	5	5	3750	75	47	49-1/2	42-1/2	49	46	46-1/2	39	8	43

TABLE II

PIPE SIZE	TYPE	PIPE / BNDL	ROWS	LAYERS	NOM. PIPE WT. BNDL	O/A DIM. A	O/A DIM. B	O/A DIM. C	MATERIALS INFO M ⚠	N ⚠	P ⚠	R	S ⚠	T	W ⚠
2	SV-DH	182	14	13	3822	71-1/2	44-1/2	46-1/4	39-3/4	45	42-1/4	42-3/4	38-3/4	24	40-1/4
2	XH-DH	156	13	12	4056	71-1/2	42-3/4	44-1/2	38-1/4	43-1/4	40-3/4	41	37	22	38-3/4
3	SV-DH	132	12	11	4092	72	51	50-3/4	46-1/4	49-1/2	48-3/4	47-1/4	44-3/4	20	46-3/4
3	XH-DH	81	9	9	3807	72	41-3/4	44-1/2	37-1/4	43-1/4	39-3/4	41	35-1/2	16	37-3/4
4	SV-DH	90	10	9	3780	72-1/2	53	51-1/4	48-1/2	50	51	47-3/4	46-1/4	16	49
4	XH-DH	64	8	8	4032	72-1/2	45-1/2	48	41	46-3/4	43-1/2	44-1/2	38-3/4	14	41-1/2
6	SV-DH	56	8	7	3808	72-1/2	59	54-3/4	54-1/2	53-1/2	57	51-1/4	51-1/4	12	55
6	XH-DH	40	8	5	4000	72-1/2	61-1/2	41-1/2	57	40-1/4	59-1/2	38	53-3/4	8	57-1/2
8	SV-DH	25	5	5	2625	75	50-1/2	52-1/2	46	52	49-3/4	50-1/4	45	8	46-1/2
8	XH-DH	25	5	5	3925	75	51-3/4	53-3/4	47-1/4	53-1/4	50-1/2	51-1/4	46	8	47-3/4

NOTES:
1. ALL DIMENSIONS IN INCHES.
2. ALL DIMENSIONS SHOWN ARE NOMINAL & WILL DIFFER DUE TO VARIATIONS IN PIPE DIAMETERS, LENGTHS, AND LUMBER THICKNESSES.
3. SV=SERVICE WT.; XH=EXTRA HEAVY; SH=SINGLE HUB; DH=DOUBLE HUB.

⚠ (1) 3 EA.–NOM. 3 X 4 X W –BOTTOM RUNNERS.
(2) 3 EA.–NOM. 2 X 4 X M –TOP RUNNERS.
(3) 6 EA.–NOM. 2 X 4 X N –UPRIGHTS.
(4) 6 EA.–NOM. 2 X 4 X P –CLOSED END BRACING.
⚠ (5) 10 EA.–NOM. 1 X 6 X R –HUB PROTECTORS TO BE DRESSED LUMBER IF REQUIRED TO STENCIL SHIPPING MARKS.
(6) T EA.–NOM. 1 X 2 X S –HORIZONTAL DIVIDERS FOR SH PACK, EXCEPT 8 INCH. LUMBER THICKNESS MUST EQUAL 7/8 INCH MINIMUM. THE TERM ROUGH SAWN LUMBER WOULD APPLY. DRESSED NOMINAL 1 INCH LUMBER WOULD NOT BE SUFFICIENT. NOM. 2 X 2 X S HORIZONTAL DIVIDERS FOR 8 INCH DH PACK (2 INCH MINIMUM THICKNESS). NOM. 1/2 X 2 X S (3/8 INCH MINIMUM THICKNESS) HORIZONTAL DIVIDERS FOR SH PACK. TWO (2) INCH WIDTH FOR ALL PACKS IS NOT CRITICAL.
(7) 1-1/4 X 0.035 - ZINC COATED STRAPPING.
(8) 6 EA.–1-1/4 -STRAPPING SEALS. (NOT ILLUSTRATED)
(9) 12 EA.–NAILS, 40d. (NOT ILLUSTRATED)
(10) 30 EA.–NAILS, 8d. (NOT ILLUSTRATED)
⚠ (11) 33 EA.–1-1/4 –STAPLES.
(12) 6 EA.–14 GA.–SAFETY WIRES.

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Figure 3. PACKING LEVEL A.

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6. NOTES

6.1 Intended use. Cast-iron pipe and fittings covered by this specification are primarily intended for use as soil pipe lines for general use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, pattern, and size of pipe and fittings required (see 1.2).
(c) Coating required if other than as specified (see 3.3).
(d) Level of packing required (see 5.1).
(e) Whether Civil agency or Military marking is required (see 5.2).

MILITARY INTERESTS:

Custodians:

Army - ME
Navy - YD
Air Force - 82

User activities:

Army - CE
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS, PCD
HEW - FEC
INTERIOR - BPA
NASA - JFK
USDA - AFS

DOD Project No. 4710-0278

PREPARING ACTIVITY: ARMY - ME

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DOCUMENT IDENTIFIER AND TITLE

WW-P-401E Pipe and Pipe Fittings, Cast-Iron, Soil

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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